CrushEval

From German brewing and more

When brewing All Grain, the quality of the crush has a major impact on the brew house efficiency as well as the ease of the lautering process.

The ideal crush of malt exposes all the endosperm as flour and grits and leaves the husks intact. But in practice this is difficult to achieve. In order to expose as much of the endosperm as possible, the malt has to be crushed very tightly which leaves most of the rather friable husk material shredded. This can lead to extremely slow run-off or stuck sparges. Conversely, when the malt is milled to loosely the husks are better preserved but a considerable amount of the endosperm remains unexposed which will result in a loss off conversion efficiency (the percentage of starch that is converted in the mash). Because of this the brewer needs to find a good compromise for setting the grain mill spacing.

Contents

What to look for

Flour to grits ratio
: the more flour the higher the efficiency but also the chance for a slow lauter or stuck sparge

Average size of the husk pieces
: the larger the husk pieces, the better they will be able to suspend the grain bed during mashing and lautering. The latter helps with the run-off speed and prevents stuck sparges.

Small husk shreds
: small husk pieces inhibit the run-off like flour.

Not or poorly crushed kernels
: If a kernel has not or only slightly been crushed, the mash may not be able to reach all the starch in that kernel and convert it into sugar or dextrins. This will lead to lower conversion efficiency.

When endosperm does remains in a crushed kernel, does it easily come out when the kernel is picked up?

There is no problem with pieces of endosperm remaining in connection with the husks. As long as they come off easily or can easily be reached by the mash water.

Examples

2 roller mill dry

36mil / 0.92mm

This crush has been produced with a 2 roller mill (JSP Maltmill) at its factory setting of 36 mil / 0.92 mm. At this setting, which is fairly loose, no substantial husk shredding occurs while the endosperm is mostly ground into grits. Some of the endosperm is left behind in the husks. The low percentage of flour and well preserved husks will make for easy lautering. But even with well modified malts, the conversion efficiency will only be average.

Malt crushed with 2 roller mill at factory setting (36 mil / 0.92 mm)(full size (http://braukaiser.com/wiki/images/6/60/Chr

close-up of the same crush(full size (http://braukaiser.com/wiki/images/e/e7/Chru

19 mil / 0.48mm

At a much tighter setting (19 mil / 0.48 mm), the same mill will produce considerably more flour and shredded husks. While the efficiency is significantly increased, the lauter may not run as easily and may even get stuck if the allowed run-off speed is to fast. The increased conversion efficiency make this an attractive option for brewers but the significantly slower run-off causes many to widen the gap at least a little bit.

Malt crushed with 2 roller mill at a tight setting (19 mil / 0.48 mm)(full size (http://braukaiser.com/wiki/images/a/a5/Crus

close-up of the same crush (full size (http://braukaiser.com/wiki/images/c/c2/Crus

2 roller mill conditioned

In order to better preserve the husk during milling, malt can be conditioned with moisture before milling. This makes the husk more pliable which results less small husk pieces even at a very tight mill setting.

36mil / 0.92mm

For this crush, the malt has been conditioned and the mill didn't tear the husks apart (only few of the husks were actually split in two halves) as it did with a dry crush. As a result some of the endosperm actually remained in the husks. But most of that endosperm actually falls right out of the husks when they are picked up and slightly rubbed between the fingers. The expected conversion efficiency should be similar to a dry crush at the same setting. In the close-up picture, the husk in the lower left corner even shows the indentation of the teeth of the malt mill roller. This illustrates how pliable the husks have become through the malt conditioning.

Conditioned malt crushed with 2 roller mill (JSP Maltmill) at factory setting (36 mil / 0.92 mm) (full size (http://braukaiser.com/wiki/images/e/e8/Crus

close-up of the same crush (full size (http://braukaiser.com/wiki/images/f/f0/Crus

19mil / 0.48mm

Even at a very tight setting of the mill, the husks of well conditioned malt remain intact while most of the endosperm is crushed into flour. There is a very small amount of husk shreds compared to a dry crush at the same. Having most of the endosperm crushed to flour will boost the conversion efficiency but also impedes lautering. Some of that slower run-off is mitigated by having the husks better preserved.

Conditioned malt crushed with 2 roller mill (JSP Maltmill) at a tight setting (19 mil / 0.48 mm) (full size (http://braukaiser.com/wiki/images/7/7b/Cru

close-up of the same crush (full size (http://braukaiser.com/wiki/images/c/c4/Crus

examples of a poor crush

food processor

This "crush" has been produced with a food processor. It is not really a crush as a food processor has spinning blades which chop the malt rather than crushing it. The results are kernels that have been broken in a way that makes it difficult for the mash to reach the endosperm which will severely limit the conversion efficiency. Rather then preserving the husks, the spinning blades produced a lot of very small husks shreds that will impede the run-off speed. A food processor is considered an unsuitable substitute for crushing malt for brewing.

malt processed with a food processor (full size (http://braukaiser.com/wiki/images/f/f1/Crus

close-up of the same "crush" (full size (http://braukaiser.com/wiki/images/a/a5/Crus

The only exception would be huskless specialty malts like Weyermann's Carafa Special malts. The latter are de-husked roasted malts. While the throughput of a food processor is good enough to process a batch the Carafa Special malts since only a small amount is generally used in a recipe.

uneven crush

This uneven crush has been produced with a two roller malt mill (JSP Maltmill) set at 90 mil / 2.3mm on one end and 12 mil / 0.30 mm on the other end. It is a poor crush as it leaves some of the kernels untouched while others are completely crushed to dust (endosperm and husks). The efficiency is expected to suffer from the not or only partially crushed kernels and the lauter will suffer from the large amount of flour that has been produced.

Malt crushed with an extremely offset 2 roller mill (full size (http://braukaiser.com/wiki/images/6/67/Cru

close-up of the same crush (full size (http://braukaiser.com/wiki/images/6/6a/Cru